Madiha M. Shahid

AI Engineer | Generative AI | LLM Engineering | RAG

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PROFESSIONAL SUMMARY

Early-career AI Engineer with professional experience progressing from AI/ML Intern to Junior AI/ML Engineer. Skilled in Python, machine learning, generative AI, LLM integration, and retrieval-augmented generation (RAG). Experienced in retraining and evaluating image-classification models, building FastAPI AI applications, implementing retrieval workflows, and integrating AI with web applications. Hands-on with LangChain, Chroma, Hugging Face, Gemini, and Google Vertex AI.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL
Machine Learning: scikit-learn, Pandas, NumPy, PyTorch, OpenCV, image classification, model evaluation
Generative AI: LLMs, prompt engineering, RAG, embeddings, structured JSON, Gemini, Groq, Qwen
AI Frameworks: LangChain, Hugging Face, Chroma, FastAPI, Django
Cloud: Google Vertex AI, Vertex AI Edge
Automation: n8n
Development: React.js, Node.js, REST APIs, Git, GitHub, API integration

PROFESSIONAL EXPERIENCE

Thriftops By SwagKicks — Karachi, Pakistan
Junior AI/ML Engineer — January 2026 – April 2026

- Integrated AI model outputs into product workflows and built React.js and Node.js screens, backed by REST APIs, for model review.
- Retrained image-classification models for product damage, components, and image angles using Google Vertex AI / Vertex AI Edge.
- Exported models for local and on-device inference and checked the 4-class damage model on local samples against pass/fail rules.

AI/ML Intern — November 2025 – December 2025

- Promoted to Junior AI/ML Engineer after this internship.
- Built a sample React.js dashboard for product and model review.
- Documented model download, local testing, and on-device inference steps for shoe and bag images on Google Vertex AI.

Elevvo — Remote
Machine Learning Intern — September 2025

- Built scikit-learn workflows in Python for cleaning, preprocessing, training, hyperparameter tuning, and evaluation.
- Handled missing values, outliers, and categorical encoding.

SELECTED PROJECTS

Support Ticket Triage Desk — *Python, LangChain, Hugging Face, Chroma, Qwen, RAG*

- Built a retrieve-then-generate RAG flow: chunk tickets and policies, embed with Hugging Face, store in Chroma, and send only the top matches to Qwen.
- Validated replies with a schema for category, priority, similar tickets, quotes, and a draft reply. Invalid JSON is rejected, and policy text overrides a conflicting ticket.

AI Decision API — *FastAPI, Gemini, RAG, JWT, SQLite, Streamlit*

- Built a ticket API that embeds store-policy markdown, retrieves the closest chunks, and returns JSON with a fixed action, confidence, reason, and source files.
- Added JWT login, per-user ticket history, retries when Gemini is rate-limited, and a script that scores decisions against sample tickets.

Job Aggregator Automation — *n8n, Groq, REST APIs, Google Sheets, Gmail*

- Built a scheduled workflow that pulls job listings, drops duplicates, asks Groq to score fit and choose a resume, logs the row, and emails only when an application address is present.

EDUCATION

B.S. Computer Science | KIET University, Karachi — Expected June 2027
CGPA 3.6/4.0